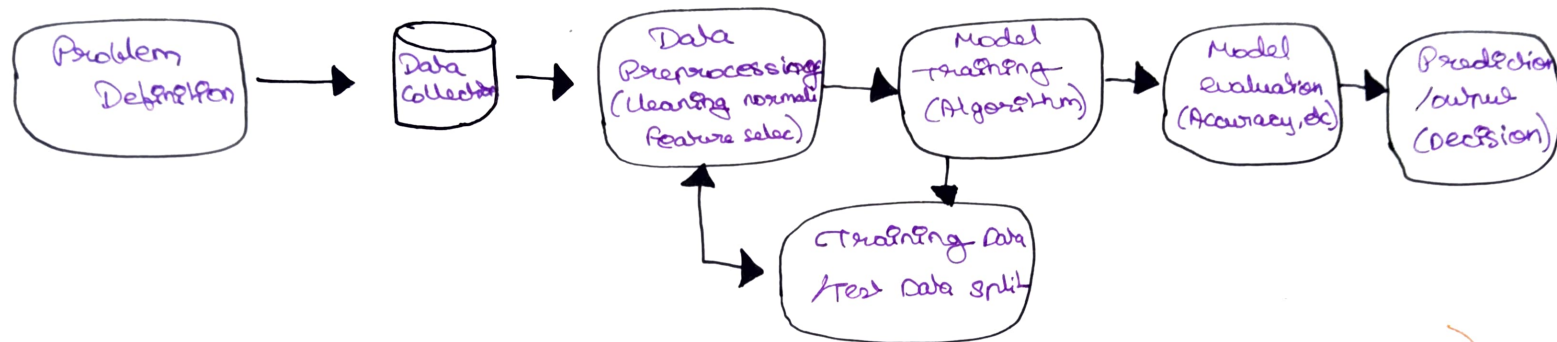


1. Blocks in a Machine Learning and Examples of Supervised & unsupervised learning



Main blocks:

1. Problem definition - define the goal (classification/prediction/clustering etc.)
2. Data collection - gather data from different sources
3. Data Preprocessing - clean data, handle missing values, normalization
4. Training/test split - divide data for training & testing
5. Model training - apply a machine learning algorithm
6. Model evaluation - check performance using metrics
7. Prediction/output - use trained model to make decision

(b) Example: For supervised learning

Task: Classify email as spam or not-spam

Input: Email Text (labeled data)

Algorithm: Decision Tree/SVM

Output: Spam or Not Spam

Example for unsupervised learning

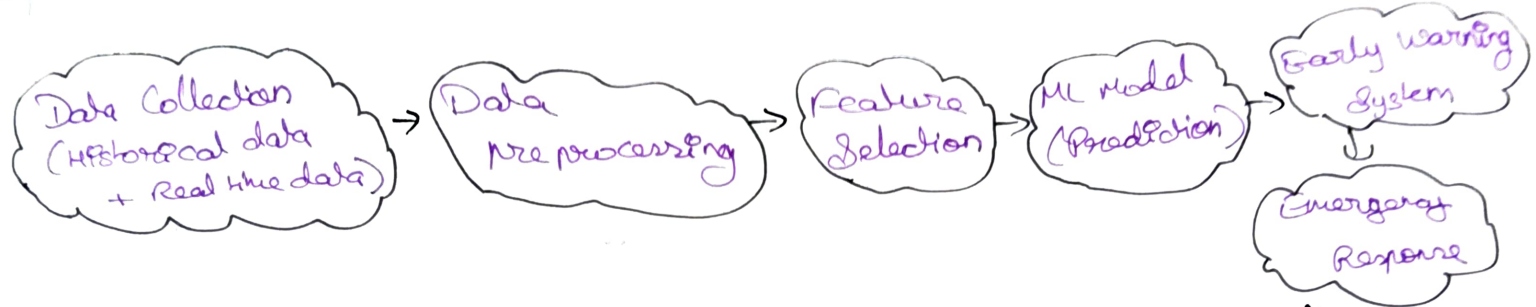
Task: Group customers into different

Input: Customer data

Algorithm: K-means clustering

Output: Different groups of customer

2. Machine Learning System for Disaster Management



Data Sources

- Historical data (past floods, earthquakes)
- Real-time data (weather sensors, seismic sensors, river water level, rainfall, satellite images)

Preprocessing:

- Handle missing values
- Noise removal
- Normalisation
- Convert categorical data
- Create time-based features

Features selection:

- Correlation analysis
- Mutual information
- Important features (e.g. rainfall, river level, soil moisture, seismic activity)

Steps for training, validating & evaluating

Data collection & preprocessing

↓
Split data

↓
Train the model

↓
validate

↓
Evaluate

Model choice

Flood prediction -
Random Forest / LSTM
(Time series)

Earthquake prediction -
Gradient Boosting /
Neural Network

How it helps early warning & emergency response

- Predicts high risk event
- Sends early alerts
- Helps in evacuation

→ Supports real time monitoring

3. Find 3 Algorithm (Play Tennis)

Concept: (Finds)

Start with the most specific hypothesis (all attributes = ϕ)

Consider only positive examples.

For each positive example, replace attribute values with "?" where they differ

Final hypothesis is the most specific hypothesis consistent with all positive examples.

Training Dataset

Ex	Outlook	Temp	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

Step by Step updates

Step	the eg	Hypothesis (S)
Initial	-	$\langle \phi, \phi, \phi, \phi \rangle$
1	$\langle \text{Overcast, Hot, High, weak} \rangle$	$\langle \text{Overcast, Hot, High, weak} \rangle$
2	$\langle \text{Rainy, Mild, High, weak} \rangle$	$\langle ?, \text{Mild, High, weak} \rangle$
3	$\langle \text{Rainy, Cool, Normal, weak} \rangle$	$\langle ?, ?, \text{Normal, weak} \rangle$
4	$\langle \text{Overcast, Mild, Normal, strong} \rangle$	$\langle ?, ?, \text{Normal, ?} \rangle$

Finally maximally specific hypothesis
 $\langle ?, ?, \text{Normal, ?} \rangle$

4. Candidate Elimination Algorithm

Concept (Candidate elimination):

Maintain two boundaries

S: most specific hypotheses

G: most general hypotheses

For a positive example: generalize S & remove inconsistent hypotheses from G

For a negative example: specialize G & remove inconsistent hypotheses from S

Final version space = all hypotheses between S & G

Training Dataset:

Ex	weather	Temp	Humidity	Wind	Enjoy spot
1	Sunny	warm	weak	.	Yes
2	Sunny	warm	strong		Yes
3	Rainy	cold	weak		No
4	Sunny	cold	weak		Yes
5	Rainy	warm	strong		No

Step by step updates

Step	Example (enjoy spot)	S set	G set
Initial	-	$\langle \phi, \phi, \phi, \phi \rangle$	$\langle ?, ?, ?, ? \rangle$
1 (+)	$\langle \text{Sunny, warm, normal, weak} \rangle$	$\langle \text{Sunny, warm, normal, weak} \rangle$	$\langle ?, ?, ?, ? \rangle$
2 (+)	$\langle \text{Sunny, warm, high, strong} \rangle$	$\langle \text{Sunny, warm, ?, ?} \rangle$	$\langle ?, ?, ?, ? \rangle$
3 (-)	$\langle \text{Rainy, cold, normal, weak} \rangle$	$\langle \text{Sunny, warm, ?, ?} \rangle$	$\langle \text{Sunny, ?, ?, ?} \rangle$

Final version space:

$S = \{ \langle \text{Sunny, ?, ?, ?} \rangle \}$

$G = \{ \langle \text{Sunny, ?, ?, ?} \rangle \}$

5. Find-S Algorithm (Customer Buying)

Concept (Find-S):

Start with the most specific hypothesis
(all attributes = ϕ).

Consider only the examples.

Replace differing attribute values with "?".

Final hypothesis is the most specific hypothesis
consistent with all the examples.

Ex	Age	Income	Student	Credit Rating	Buys Product
1	Middle	High	No	Fair	Yes
2	Senior	Medium	No	Fair	Yes
3	Junior	Low	Yes	Fair	Yes

Step by step updates

Step	The example	Hypothesis
Initial	-	$\langle \phi, \phi, \phi, \phi \rangle$
1	$\langle \text{Middle}, \text{High}, \text{No}, \text{Fair} \rangle$	$\langle \text{Middle}, \text{High}, \text{No}, \text{Fair} \rangle$
2	$\langle \text{Senior}, \text{Medium}, \text{No}, \text{Fair} \rangle$	$\langle ?, ?, \text{No}, \text{Fair} \rangle$
3	$\langle \text{Junior}, \text{Low}, \text{Yes}, \text{Fair} \rangle$	$\langle ?, ?, ?, \text{Fair} \rangle$

Final Maximally specific hypothesis

$\langle ?, ?, ? \text{Fair} \rangle$